Stat 175: Homework 3

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Please submit your solutions to Gradescope by **Friday, February 20th, 11:59pm**.

Exercise 1 (SBM Communities). Let $G = (V, E)$ be a random graph on n vertices from a Stochastic Block Model (SBM) with $k = 2$ communities. Each vertex i belongs to community $z_i \in \{1, 2\}$ with equal probability: $\pi_1 = \pi_2 = \frac{1}{2}$. Conditioned on the community labels $(z_i)_{i \in V}$, the edges are independent and satisfy

$$\mathbb{P}\left((i,j) \in E \mid z_i = a, z_j = b\right) = \begin{cases} p, & a = b \\ q, & a \neq b. \end{cases}$$

for some $0 \leq p, q \leq 1$. Thus p is the within-community connection probability, and q is the between-community connection probability. For two distinct vertices i and j, define the number of common neighbors

$$C_{ij} = \left|\{k \in V \setminus \{i,j\} : k \text{ is adjacent to both } i, j\}\right|.$$

(a) Compute $\mathbb{E}[C_{ij} \mid z_i = z_j]$, the expected number of common neighbors of i and j when they are in the *same* community.

(b) Compute $\mathbb{E}[C_{ij} \mid z_i \neq z_j]$, the expected number of common neighbors of i and j when they are in *different* communities.

(c) Compare your two expressions. Which of the two expectations is larger (or are they equal)? Does your conclusion depend on whether $p > q$ or $p < q$?

(d) Based on your answers, propose a simple rule that attempts to decide whether a given pair (i, j) belongs to the same community or to different communities.

Solution to Exercise 1

Let $V = \{1, \ldots, n\}$ and fix distinct $i \neq j$. For each $k \in V \setminus \{i, j\}$ define the indicator

$$I_k := \mathbf{1}\{k \text{ adjacent to both } i \text{ and } j\}.$$

Then

$$C_{ij} = \sum_{k \neq i,j} I_k, \qquad \mathbb{E}[C_{ij} \mid \cdot] = \sum_{k \neq i,j} \mathbb{E}[I_k \mid \cdot].$$

Conditioned on community labels, edges are independent, so

$$\mathbb{E}[I_k \mid z_i, z_j, z_k] = \mathbb{P}((i,k) \in E \mid z_i, z_k) \cdot \mathbb{P}((j,k) \in E \mid z_j, z_k).$$

(a) Assume $z_i = z_j$. Write $a := z_i = z_j \in \{1, 2\}$. For a given k,

$$\mathbb{E}[I_k \mid z_i = z_j = a] = \mathbb{E}\Big[\mathbb{P}(i \sim k \mid a, z_k)\mathbb{P}(j \sim k \mid a, z_k) \,\Big|\, z_i = z_j = a\Big].$$

Now $z_k = a$ with probability $1/2$ and $z_k \neq a$ with probability $1/2$.

- If $z_k = a$, then $\mathbb{P}(i \sim k) = p$ and $\mathbb{P}(j \sim k) = p$, so the product is p^2.
- If $z_k \neq a$, then $\mathbb{P}(i \sim k) = q$ and $\mathbb{P}(j \sim k) = q$, so the product is q^2.

Hence

$$\mathbb{E}[I_k \mid z_i = z_j] = \frac{1}{2}p^2 + \frac{1}{2}q^2.$$

Summing over the $n-2$ possible k gives

$$\boxed{\mathbb{E}[C_{ij} \mid z_i = z_j] = (n-2) \cdot \frac{p^2+q^2}{2}}.$$

(b) Assume $z_i \neq z_j$. Without loss of generality take $z_i = 1, z_j = 2$. For a given k, we again average over z_k:

- If $z_k = 1$ (probability $1/2$), then i and k are same-community so $\mathbb{P}(i \sim k) = p$, while j and k are different-community so $\mathbb{P}(j \sim k) = q$. Product: pq.
- If $z_k = 2$ (probability $1/2$), then $\mathbb{P}(i \sim k) = q$ and $\mathbb{P}(j \sim k) = p$. Product: $qp = pq$.

So in either case the product is pq, hence

$$\mathbb{E}[I_k \mid z_i \neq z_j] = pq,$$

and therefore

$$\boxed{\mathbb{E}[C_{ij} \mid z_i \neq z_j] = (n-2)\,pq}.$$

(c) Compare:

$$\mathbb{E}[C_{ij} \mid z_i = z_j] - \mathbb{E}[C_{ij} \mid z_i \neq z_j] = (n-2)\left(\frac{p^2+q^2}{2} - pq\right) = (n-2) \cdot \frac{(p-q)^2}{2} \geq 0.$$

Thus

$$\boxed{\mathbb{E}[C_{ij} \mid z_i = z_j] \geq \mathbb{E}[C_{ij} \mid z_i \neq z_j] \text{ always, with equality iff } p = q.}$$

So the "same-community" expectation is larger regardless of whether $p > q$ or $p < q$; it only depends on $p \neq q$.

(d) A simple decision rule: for an observed pair (i, j), compute C_{ij}. Since the expected number of common neighbors is larger when i, j are in the same community (when $p \neq q$), a natural classifier is a threshold test:

$$\text{declare } z_i = z_j \text{ if } C_{ij} \text{ is "large," and } z_i \neq z_j \text{ otherwise.}$$

One concrete choice is to compare C_{ij} to the midpoint of the two conditional means:

$$T := \frac{1}{2}\Big(\mathbb{E}[C_{ij} \mid z_i = z_j] + \mathbb{E}[C_{ij} \mid z_i \neq z_j]\Big) = (n-2)\left(\frac{p^2+q^2}{4} + \frac{pq}{2}\right).$$

Then:

$$\boxed{\text{Predict same community if } C_{ij} > T, \text{ predict different if } C_{ij} < T \text{ (ties arbitrary).}}$$

(Any monotone threshold rule in C_{ij} follows the same intuition.)

Exercise 2 (Contextualized SBM). In the Contextualized Stochastic Block Model (CSBM), we observe both a graph G and a set of node features X. Let n be the number of nodes and $k = 2$ be the number of communities. Each node i has a latent community $y_i \in \{+1, -1\}$ assigned uniformly at random. Let $\boldsymbol{v} \sim N(0, \frac{1}{d}\mathbf{I}_d)$. We associate each node i with a feature vector

$$X_i = \sqrt{\frac{1}{d}} y_i \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|} + \sigma \mathbf{Z}_i,$$

where $\sigma > 0$ and $\mathbf{Z}_i \sim N(0, \frac{1}{d}\mathbf{I}_d)$. Suppose again that the within-community edge probability is $p_{11} = p_{-1,-1} = p$ and the between-community edge probability is $p_{1,-1} = q$. Let $n = 500$ and $d = 250$.

(*a*) First, let us ignore the features (X_i) for now, and focus only on the graph G. Recall the two main steps of the Spectral Clustering Algorithm mentioned in class:

1. Compute the eigenvector e_2 which corresponds to the second smallest eigenvalue λ_2 of the Laplacian matrix $L = D - A$ (recall that we already know $\lambda_1 = 0$, and $e_1 = (1, \ldots, 1)$).
2. Cluster the entries $(e_{2,i})_{i \le n}$ using the k-means algorithm, and assign communities from this.

Do the above algorithm for three different sets of values for p and q:

$$\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix}, \quad \begin{pmatrix} \frac{4\log^2(n)}{3n} \\ \frac{\log^2(n)}{n} \end{pmatrix}, \quad \begin{pmatrix} \frac{4}{3n} \\ \frac{1}{n} \end{pmatrix}.$$

What do you observe?

(*b*) Imagine now that $\sigma = 0$, meaning $X_i = \sqrt{\frac{1}{d}} y_i \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|}$. What is the top eigenvector of $\frac{1}{d}XX^\intercal$? If σ is instead small but not exactly zero, what can we expect for the top eigenvector of $\frac{1}{d}XX^\intercal$?

(*c*) Implement the following spectral method: obtain the top eigenvector of $\frac{1}{d}XX^\intercal$, and call it $\hat{v}_1$. Run the k-means clustering algorithm on the entries of $\hat{v}_1$, and assign communities correspondingly. Do this for $\sigma = 0, 0.1, 0.5, 1$. What do you observe?

(*d*) Now, for each vertex i, define the vector $B_i := (e_{2,i}, \hat{v}_{1,i})$. Run the k-means algorithm on $(B_i)_{i \le n}$ and classify into communities again. What do you observe now?

Solution to Exercise 2

(a) Figure **??** shows the entries of e_2 (sorted by true label) for each (p, q) regime.

- **Dense regime** $(p, q) = (0.8, 0.6)$: The two communities are perfectly separated by e_2; k-means recovers them exactly.
- **Logarithmic regime** $p = \frac{4\log^2 n}{3n}$, $q = \frac{\log^2 n}{n}$: There is a clear sign flip in e_2 at the community boundary; clustering is again accurate.
- **Subcritical regime** $p = \frac{4}{3n}$, $q = \frac{1}{n}$: The graph is very sparse and the eigenvector entries are essentially noise — no community structure is visible and k-means fails.

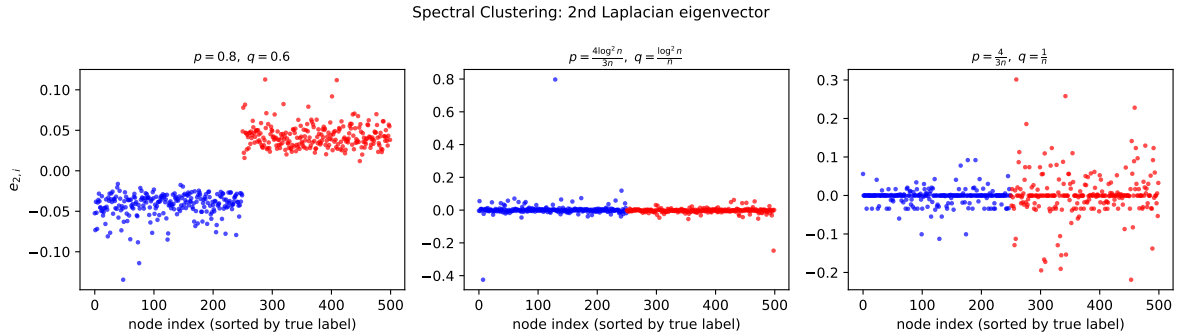


Figure 1: Second eigenvector e_2 of $L = D - A$, entries coloured by true label ($+1$ red, -1 blue).

(b) **Top eigenvector of $\frac{1}{d}XX^\top$ for $\sigma = 0$, and small σ.**

Let $y = (y_1, \ldots, y_n)^\top \in \{\pm 1\}^n$ and define the unit direction $u := \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|} \in \mathbb{R}^d$ (well-defined a.s.). When $\sigma = 0$, each feature vector is

$$X_i = \sqrt{\frac{1}{d}}\, y_i\, u.$$

Stacking rows gives an $n \times d$ matrix X of the form

$$X = \sqrt{\frac{1}{d}}\, y\, u^\top.$$

Then

$$XX^\top = \frac{1}{d}\, y\, u^\top u\, y^\top.$$

Since $\|u\| = 1$, we have $u^\top u = 1$, so

$$XX^\top = \frac{1}{d}\, yy^\top \qquad \Rightarrow \qquad \frac{1}{d}XX^\top = \frac{1}{d^2}\, yy^\top.$$

This is a rank-one matrix. Its unique (up to sign) top eigenvector is proportional to y:

$$yy^\top y = (y^\top y)\, y = n\, y,$$

so the eigenvalue is $\frac{n}{d^2}$ and any unit-norm top eigenvector is

$$\boxed{v_1 = \pm\frac{y}{\|y\|} = \pm\frac{y}{\sqrt{n}}}.$$

Therefore, clustering the entries of v_1 perfectly recovers the two communities (up to label swap) when $\sigma = 0$.

If $\sigma > 0$ but small, write

$$X = \sqrt{\frac{1}{d}}\, yu^\top + \sigma Z,$$

where Z has independent rows $Z_i \sim N(0, \frac{1}{d} I_d)$. Then

$$\frac{1}{d}XX^\top = \underbrace{\frac{1}{d^2}\, yy^\top}_{\text{rank-1 signal}} + \underbrace{\frac{\sigma}{d}\left(\sqrt{\frac{1}{d}}yu^\top Z^\top + \sqrt{\frac{1}{d}}Zuy^\top\right)}_{\text{cross terms}} + \underbrace{\frac{\sigma^2}{d}ZZ^\top}_{\text{noise}}.$$

Heuristically, this is a *spiked* random matrix: a rank-one signal $yy^\top$ plus an approximately isotropic noise term. When σ is small enough (signal-to-noise large enough), standard perturbation intuition (e.g. Davis–Kahan) says the top eigenvector remains *close* to the signal direction y:

$$\boxed{\text{For small } \sigma,\ \hat{v}_1 \text{ is strongly correlated with } y \text{ (close to } \pm y/\sqrt{n}).}$$

As σ increases, the top eigenvector becomes less aligned with y and eventually can be dominated by the noise.

(c) Figure **??** shows the entries of the top eigenvector $\hat{v}_1$ of $\frac{1}{d}XX^\top$ for each σ.

- $\sigma = 0$: $\hat{v}_1$ is exactly proportional to the label vector y; perfect clustering.
- $\sigma = 0.1$: Still a clear two-block structure; near-perfect accuracy.
- $\sigma = 0.5$: The signal is weaker but still visible; clustering succeeds with moderate error.
- $\sigma = 1$: Noise dominates; the entries no longer separate into two groups and k-means largely fails.

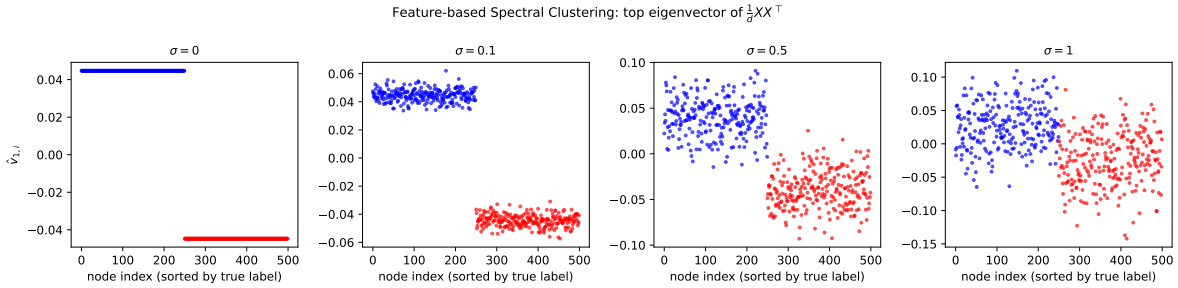


Figure 2: Top eigenvector $\hat{v}_1$ of $\frac{1}{d}XX^\top$, entries coloured by true label, for $\sigma \in \{0, 0.1, 0.5, 1\}$.

(d) Using the graph from the logarithmic (p, q) regime (part (a)) and features with $\sigma = 0.5$ (part (c)), we form $B_i = (e_{2,i}, \hat{v}_{1,i})$ and run k-means on these 2-D vectors. Figure **??** shows the scatter plot of B_i coloured by whether each node is correctly classified. Combining both sources of information yields better accuracy than either alone at $\sigma = 0.5$: the graph eigenvector and the feature eigenvector each carry partial signal, and their union provides a cleaner separation boundary.

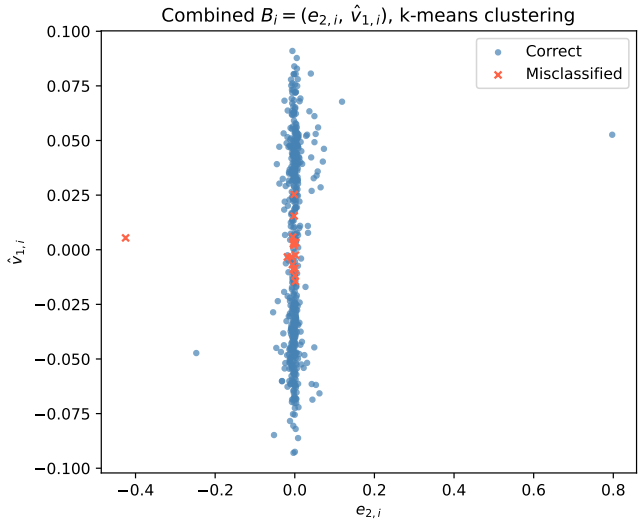


Figure 3: Scatter of $B_i = (e_{2,i}, \hat{v}_{1,i})$; blue = correctly classified, red $\times$ = misclassified.

Exercise 3 (Motifs in Erdős–Rényi). Consider the Erdős–Rényi Random Graph $G(n,p)$ on vertices $\{1,\dots,n\}$, where each edge exists independently with probability p. Let H be a fixed simple graph (our motif) with vertex set $V(H)$ and edge set $E(H)$, and define

$$v(H) := |V(H)| \qquad e(H) := |E(H)| \qquad d(H) := \frac{e(H)}{v(H)}.$$

Let X be the number of (labelled) copies of H inside our $G(n,p)$ graph.

(a) Show that

$$\mathbb{E}[X] = \Theta\left(n^{v(H)}p^{e(H)}\right) \quad \text{as} \quad n \to \infty,$$

where the implicit constants may depend on H, but not on n or p.

(b) Deduce that if

$$p \ll n^{-1/d(H)} \quad \left(\text{i.e., } p \cdot n^{1/d(H)} \to 0\right),$$

then we must have $\mathbb{E}[X] \to 0$ as $n \to \infty$.

(c) Conclude that, under this condition on p, the probability that $G(n,p)$ contains at least one copy of H also tends to 0 as $n \to \infty$.

Solution to Exercise 3

Let $v := v(H) = |V(H)|$ and $e := e(H) = |E(H)|$. A *labelled copy* of H in $G(n,p)$ means an injective map

$$\varphi : V(H) \to [n] := \{1,\dots,n\}$$

such that for every edge $\{a,b\} \in E(H)$, the image $\{\varphi(a),\varphi(b)\}$ is an edge in $G(n,p)$. (Since H is simple, this is the natural notion.)
For each injection φ, define the indicator random variable

$$I_\varphi := \mathbf{1}\{\varphi \text{ is a graph embedding of } H \text{ into } G(n,p)\}.$$

Then

$$X = \sum_{\varphi \text{ injective}} I_\varphi \qquad \Rightarrow \qquad \mathbb{E}[X] = \sum_{\varphi \text{ injective}} \mathbb{E}[I_\varphi].$$

(a) For a fixed injection φ, the event $\{I_\varphi = 1\}$ requires that all e edges of H appear among the corresponding pairs in $G(n,p)$. In $G(n,p)$, edges appear independently with probability p, so

$$\mathbb{E}[I_\varphi] = \mathbb{P}(I_\varphi = 1) = p^e.$$

It remains to count the number of injective maps $\varphi : V(H) \to [n]$, which is the falling factorial

$$(n)_v := n(n-1)\cdots(n-v+1).$$

Therefore

$$\mathbb{E}[X] = (n)_v\, p^e.$$

Since H is fixed (so v is fixed) and $n \to \infty$, we have the standard asymptotic

$$(n)_v = n^v\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{v-1}{n}\right) = \Theta(n^v),$$

with implicit constants depending only on v (hence only on H). Hence

$$\boxed{\mathbb{E}[X] = \Theta(n^{v(H)}p^{e(H)}) \text{ as } n \to \infty.}$$

(b) Recall $d(H) = e(H)/v(H) = e/v$. If $p \ll n^{-1/d(H)}$, then $p\, n^{1/d(H)} \to 0$, i.e.

$$p\, n^{v/e} \to 0.$$

Raise both sides to the power e (note e is fixed):

$$p^e\, n^v \to 0.$$

Using part (a), $\mathbb{E}[X] = \Theta(n^v p^e)$, so it follows that

$$\boxed{\mathbb{E}[X] \to 0 \text{ as } n \to \infty.}$$

(c) Let A be the event that $G(n,p)$ contains at least one copy of H. Then $A = \{X \geq 1\}$. By Markov's inequality,

$$\mathbb{P}(X \geq 1) \leq \mathbb{E}[X].$$

Under the condition in (b), $\mathbb{E}[X] \to 0$, hence

$$\boxed{\mathbb{P}(G(n,p) \text{ contains at least one copy of } H) = \mathbb{P}(X \geq 1) \to 0.}$$